

# Dihedral

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# Chapter 1

## Introduction

This blueprint follows the paper *Combinatorial curve neighborhoods for the affine flag manifold of type  $A_1^{(1)}$* . The formalization works in the infinite dihedral group  $D_\infty$  and proves the paper's main combinatorial formula

$$\Gamma_d(u) = \{uw \mid w \in \max A_d(u)\}.$$

The Lean development keeps the paper-level objects visible: chains in the moment graph, the degree map  $\varphi$ , the auxiliary set  $A_d(u)$ , the maximal root  $\alpha(d)$ , and the curve neighborhood  $\Gamma_d(u)$ .

## Chapter 2

# Section 2A: The Infinite Dihedral Group

**Theorem 1** (Generators). *The group  $D_\infty$  is generated by the two simple reflections  $sr(0)$  and  $sr(1)$ .*

**Definition 2** (Length). Let  $\ell : D_\infty \rightarrow \mathbb{N}$  be the Coxeter length.

**Theorem 3** (Reduced-word length). *For every  $g \in D_\infty$ ,*

$$\ell(g) = \text{length}(\text{reducedWord}(g)).$$

**Theorem 4** (Reflection length). *For every  $k \in \mathbb{Z}$ ,*

$$\ell(sr(k)) = |2k - 1|.$$

**Lemma 5** (Length dichotomy). *If  $\ell(u) \leq \ell(v)$ , then*

$$\ell(uv) = \ell(u) + \ell(v) \quad \text{or} \quad \ell(uv) = \ell(v) - \ell(u).$$

**Definition 6** (Roots and root reflections). A positive root is a pair  $(a, b) \in \mathbb{N}^2$  with  $a = b + 1$  or  $b = a + 1$ . Its root reflection is the alternating word  $s_\alpha$  of length  $a + b$ , and Lean proves  $\ell(s_\alpha) = a + b$  and  $\varphi(s_\alpha) = \alpha$ .

**Definition 7** (Degree of an element). A degree is a pair of natural numbers with product order. The map  $\varphi : D_\infty \rightarrow \mathbb{N}^2$  records the numbers of  $s_0$  and  $s_1$  in the reduced word.

## Chapter 3

# Section 2B: Moment Graph, Chains, and Order

**Definition 8** (Edges and chains). An edge  $u \xrightarrow{\alpha} v$  means  $v = us_{\alpha}$ . A chain is a finite succession of such edges; its degree is the sum of the root degrees on the edges.

**Theorem 9** (Bruhat order by length). *For vertices  $u, v$ ,*

$$u < v \iff \ell(u) < \ell(v).$$

**Lemma 10** (Paper Lemma 2.5). *If there is a chain from  $u$  to  $v$  of total degree  $d$ , then*

$$d = \varphi(u^{-1}v) + 2(r, s)$$

*for some  $r, s \in \mathbb{N}$ . In particular  $\varphi(u^{-1}v) \leq d$ .*

**Definition 11** (Reachable set and curve neighborhood). The reachable set is

$$\text{ReachableSet}(u, d) = \{v \mid \exists d' \leq d, \text{HasChain } u \ v \ d'\}.$$

The curve neighborhood  $\Gamma_d(u)$  is the set of maximal elements of this reachable set.

## Chapter 4

# Section 3: Calculation of Curve Neighborhoods

**Definition 12** (The set  $A_d(u)$ ). For  $u \in D_\infty$  and degree bound  $d$ ,

$$A_d(u) = \{v \mid \ell(uv) = \ell(u) + \ell(v), \varphi(v) \leq d\}.$$

**Definition 13** (The maximal root  $\alpha(d)$ ). For non-diagonal  $d$ ,  $\alpha(d)$  is the maximal root below  $d$ :

$$\alpha(d) = \begin{cases} (d.a, d.a + 1), & d.a < d.b, \\ (d.b + 1, d.b), & d.a > d.b. \end{cases}$$

Lean keeps the old names `root_from_degree` and `s_alpha_d`, and exposes the semantic aliases `maxRootLE` and `rootReflectionLE`.

**Lemma 14** (Paper Lemma 3.1). *The set  $A_d(u)$  has a unique maximal element unless  $u = 1$  and  $d = (a, a)$ . In the exceptional diagonal identity case, the maximal elements are  $r(a)$  and  $r(-a)$ .*

**Theorem 15** (Formula (3): identity base point). *At the identity,*

$$\Gamma_d(1) = \begin{cases} \{(s_0 s_1)^a, (s_1 s_0)^a\}, & d = (a, a), \\ \{s_{\alpha(d)}\}, & d.a \neq d.b. \end{cases}$$

**Theorem 16** (Formula (4): one endpoint case). *If the reduced word for  $u$  ends in  $s_0$ , Lean defines the paper's candidate maximal element by `maxAdCandidateEndsS0 d`. The theorem `max_ad_candidate_ends_s0` proves that this candidate is maximal in  $A_d(u)$ . The case ending in  $s_1$  is symmetric and is not duplicated as a separate public formula.*

**Lemma 17** (Paper Lemma 3.4). *If  $z \in A_d(1)$  and  $v \in \Gamma_d(u)$ , then  $\ell(uz) \leq \ell(v)$  and  $\varphi(u^{-1}v) \leq d$ . If moreover  $z \in \Gamma_d(1)$ , then  $\ell(u^{-1}v) \leq \ell(z)$ .*

**Lemma 18** (Paper Lemma 3.5). *If  $v \in \Gamma_d(u)$ , then*

$$u^{-1}v \in A_d(u).$$

**Theorem 19** (Paper Theorem 3.6). *For every  $u \in D_\infty$  and every degree bound  $d$ ,*

$$\Gamma_d(u) = \{v \mid \exists w, w \in \max A_d(u) \text{ and } v = uw\}.$$

## Chapter 5

# Computable Appendix

**Definition 20** (Finite enumeration). The computable model enumerates all elements up to the finite length bound  $d.a + d.b + 1$  and filters by the defining conditions of  $A_d(u)$ .

**Theorem 21** (Finite model correctness). *For every base point  $u$ , degree bound  $d$ , and vertex  $v$ ,*

$$v \in \textit{CurveNeighborhood\_computable } u \ d \iff v \in \Gamma_d(u).$$

## Chapter 6

# Local Support Lemmas

**Lemma 22** (Integer and length API). *The local support files stage the integer absolute-value and rank-two length facts needed to keep the infinite-dihedral model stable under mathlib changes.*